

Project Blueprint

Federated Intrusion Detection System with Poisoning-Resilient Aggregation

Team:

- **Team Member 1** — Federated learning system build & integration
 - **Team Member 2** — Data analytics, evaluation & benchmarking
 - **Team Member 3** — Security learning, attack simulation
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STAGE 1 — Semester 1: Build the Foundation

Checkpoint	Timeline	What We'll Deliver	Owner
Checkpoint 1 Team Ready	Week 4	Team has learned the core concepts (FL, IDS, security basics). Dataset (CICIDS2017) acquired and understood.	All
Checkpoint 2 Data Pipeline Ready	Week 6	Clean, feature-engineered dataset ready for training, with exploratory analysis.	Team Member 2
Checkpoint 3 Centralized Baseline	Week 8	A standard (non-federated) intrusion detection model, trained and evaluated. This is our benchmark.	Team Member 1
Checkpoint 4	Week 12	Same intrusion detection task, now	Team Member 1 & 3

Checkpoint	Timeline	What We'll Deliver	Owner
Federated System Working		trained collaboratively across simulated organizations without sharing raw data.	
Checkpoint 5 Stage 1 Complete	Week 14	Full comparison: centralized vs. federated performance. Report + presentation.	Team Member 2

End of Stage 1, we will have: A working federated intrusion detection system that performs comparably to a traditional centralized system — proving collaborative, privacy-preserving detection is viable.

STAGE 2 — Semester 2: Make It Adversarially Robust

Checkpoint	Timeline	What We'll Deliver	Owner
Checkpoint 6 Threat Model Defined	Week 3	Understanding of how malicious participants could sabotage the system, and which defenses we'll use.	Team Member 3
Checkpoint 7 Vulnerability Proven	Week 6	Demonstration that the Stage 1 system fails/degrades when malicious participants are introduced (no defense yet).	Team Member 3
Checkpoint 8 Defense Implemented	Week 9	Poisoning-resilient version of the system, using robust aggregation to detect and neutralize	Team Member 1

Checkpoint	Timeline	What We'll Deliver	Owner
		malicious participants.	
Checkpoint 9 Full Evaluation	Week 12	Complete 3-way benchmark: centralized vs. undefended federated vs. defended federated, across varying attack severity.	Team Member 2
Checkpoint 10 Final Delivery	Week 15	Final report, working demo, and presentation.	All

End of Stage 2, we will have: A federated intrusion detection system that stays reliable even when some participants are actively malicious — with proof, not just claims.

One-Line Summary Per Stage

- **Stage 1:** Build a working, privacy-preserving intrusion detection system across multiple organizations.
- **Stage 2:** Prove it stays secure even when one of those organizations turns malicious, and fix it if it doesn't.